

Homework Problems. *Solutions should be written/typed in numerical order and clearly labeled. Each page should have two problems (except the last page if there are an odd number of problems). Unsolved problems should be included but left blank.*

1. Write in tabular form:

- a. $A = \{x \mid x \text{ is a letter in "correct"}\}$
- b. $B = \{x \mid x \text{ is positive and } x \text{ is negative}\}$
- c. $C = \{x \in \mathbb{Z} \mid x - 2 \leq 5\}$

2. Write these sets in a set-builder form:

- a. Let $A = \{3, 4, 5, 6, 7, \dots\}$
- b. Let $B = \{3, 5, 7, 9, \dots\}$
- c. Let C be the US presidents elected after 1999.

3. Find all the subsets of $A = \{x, y, z\}$.

4. Prove that every nonempty subset has a proper subset.

5. Prove that if A is a subset of $\emptyset$, then $A = \emptyset$.

6. Prove that $A = \{2, 3, 4, 5\}$ is not a subset of $B = \{x \mid x \text{ is even}\}$.

7. Prove that A and B are subsets of $A \cup B$.

8. Prove that $A = A \cup A$ and $A = A \cap A$.

9. Prove that $A \cap B$ is a subset of A and of B .

10. Prove that $A \cup B = \emptyset$ implies that $A = \emptyset$ and $B = \emptyset$.

Presentation problems.

1. Prove that $A - B$ is a subset of $A \cup B$.

2. Prove that if A is a subset of B then $A \cap B = A$.

3. Let $A \cap B = \emptyset$. Prove that $B \cap A' = B$.

4. Prove that if A is a subset of B then $A \cup B = B$.

5. Prove that $A' - B' = B - A$.

6. Prove that if A is a subset of B then B' is a subset of A' .

7. Prove that if $A \cap B = \emptyset$ then $A \cup B' = B'$.

8. Prove that $(A \cap B)' = A' \cup B'$.

9. Prove that $A \subset B$ implies that $A \cup (B - A) = B$.